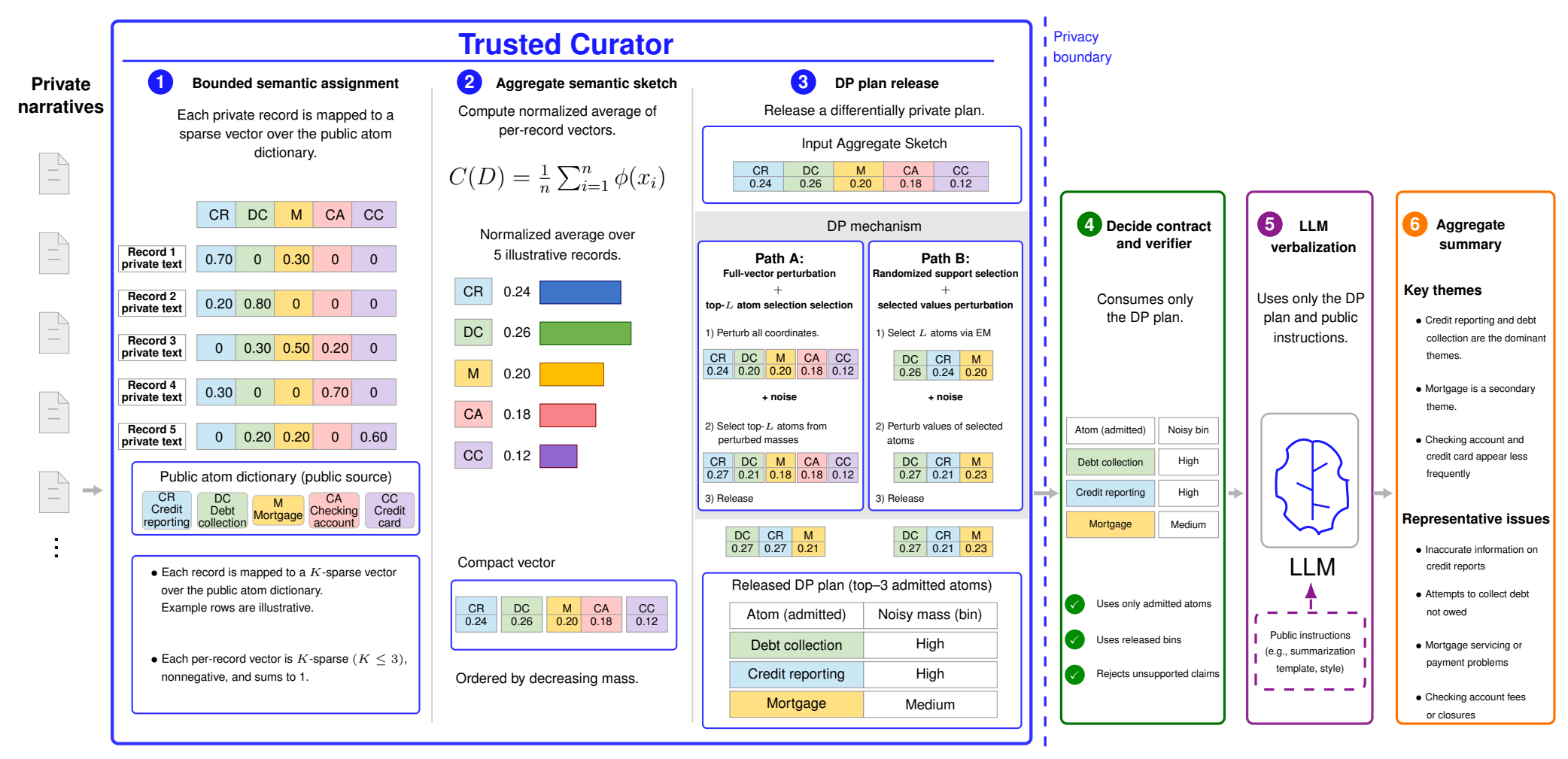

Private narratives
Trusted Curator
1 Bounded semantic assignment
Each private record is mapped to a sparse vector over the public atom dictionary.
CR DC M CA CC
Record 1 private text 0.70 0 0.30 0 0
Record 2 private text 0.20 0.80 0 0 0
Record 3 private text 0 0.30 0.50 0.20 0
Record 4 private text 0.30 0 0 0.70 0
Record 5 private text 0 0.20 0.20 0 0.60
Public atom dictionary (public source)
CR Credit reporting
DC Debt collection
M Mortgage
CA Checking account
CC Credit card
• Each record is mapped to a K-sparse vector over the public atom dictionary. Example rows are illustrative.
• Each per-record vector is K-sparse ($K \leq 3$), nonnegative, and sums to 1.
2 Aggregate semantic sketch
Compute normalized average of per-record vectors.
$C(D) = \frac{1}{n}\sum_{i=1}^{n}\phi(x_i)$
Normalized average over 5 illustrative records.
CR 0.24
DC 0.26
M 0.20
CA 0.18
CC 0.12
Compact vector
CR 0.24 DC 0.26 M 0.20 CA 0.18 CC 0.12
Ordered by decreasing mass.
3 DP plan release
Release a differentially private plan.
Input Aggregate Sketch
CR 0.24 DC 0.26 M 0.20 CA 0.18 CC 0.12
DP mechanism
Path A:
Full-vector perturbation
+
top-L atom selection selection
1) Perturb all coordinates.
CR 0.24 DC 0.20 M 0.20 CA 0.18 CC 0.12
+ noise
2) Select top-L atoms from perturbed masses
CR 0.27 DC 0.21 M 0.18 CA 0.18 CC 0.12
3) Release
DC 0.27 CR 0.27 M 0.21
Path B:
Randomized support selection
+
selected values perturbation
1) Select L atoms via EM
DC 0.26 CR 0.24 M 0.20
+ noise
2) Perturb values of selected atoms
DC 0.27 CR 0.21 M 0.23
3) Release
DC 0.27 CR 0.21 M 0.23
Released DP plan (top–3 admitted atoms)
Atom (admitted) Noisy mass (bin)
Debt collection High
Credit reporting High
Mortgage Medium
Privacy boundary
4 Decide contract and verifier
Consumes only the DP plan.
Atom (admitted) Noisy bin
Debt collection High
Credit reporting High
Mortgage Medium
Uses only admitted atoms
Uses released bins
Rejects unsupported claims
5 LLM verbalization
Uses only the DP plan and public instructions.
LLM
Public instructions (e.g., summarization template, style)
6 Aggregate summary
Key themes
• Credit reporting and debt collection are the dominant themes.
• Mortgage is a secondary theme.
• Checking account and credit card appear less frequently
Representative issues
• Inaccurate information on credit reports
• Attempts to collect debt not owed
• Mortgage servicing or payment problems
• Checking account fees or closures